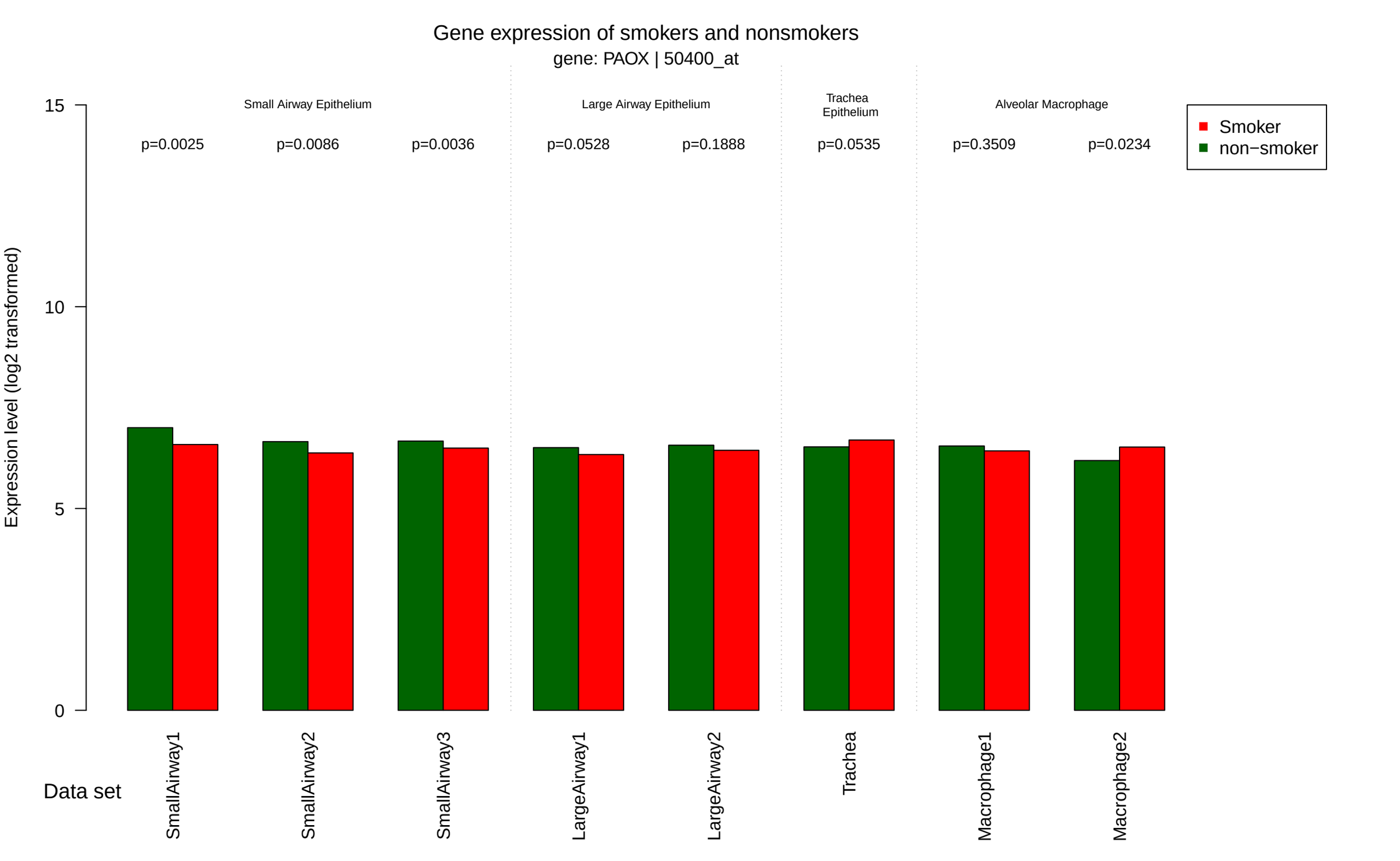
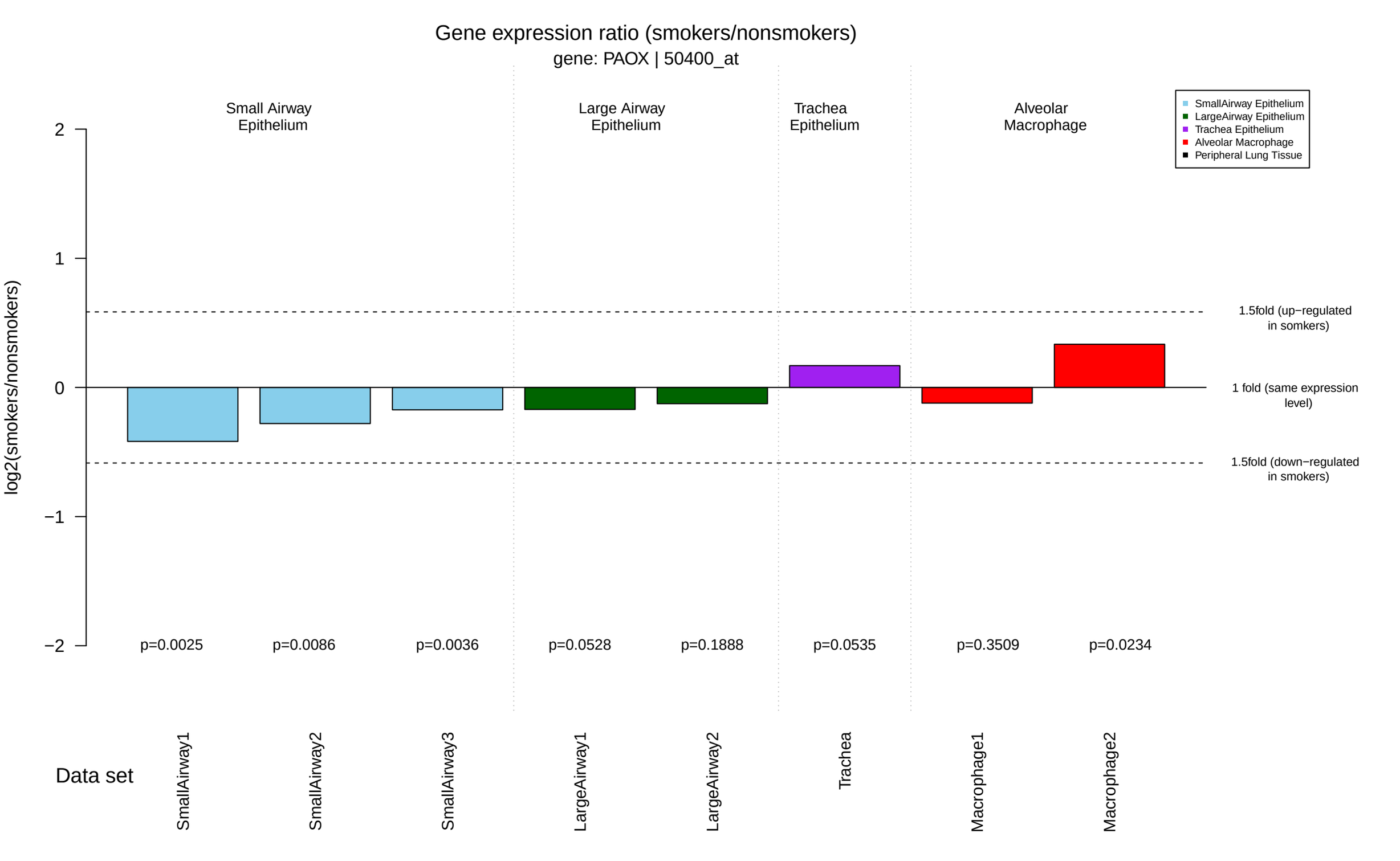
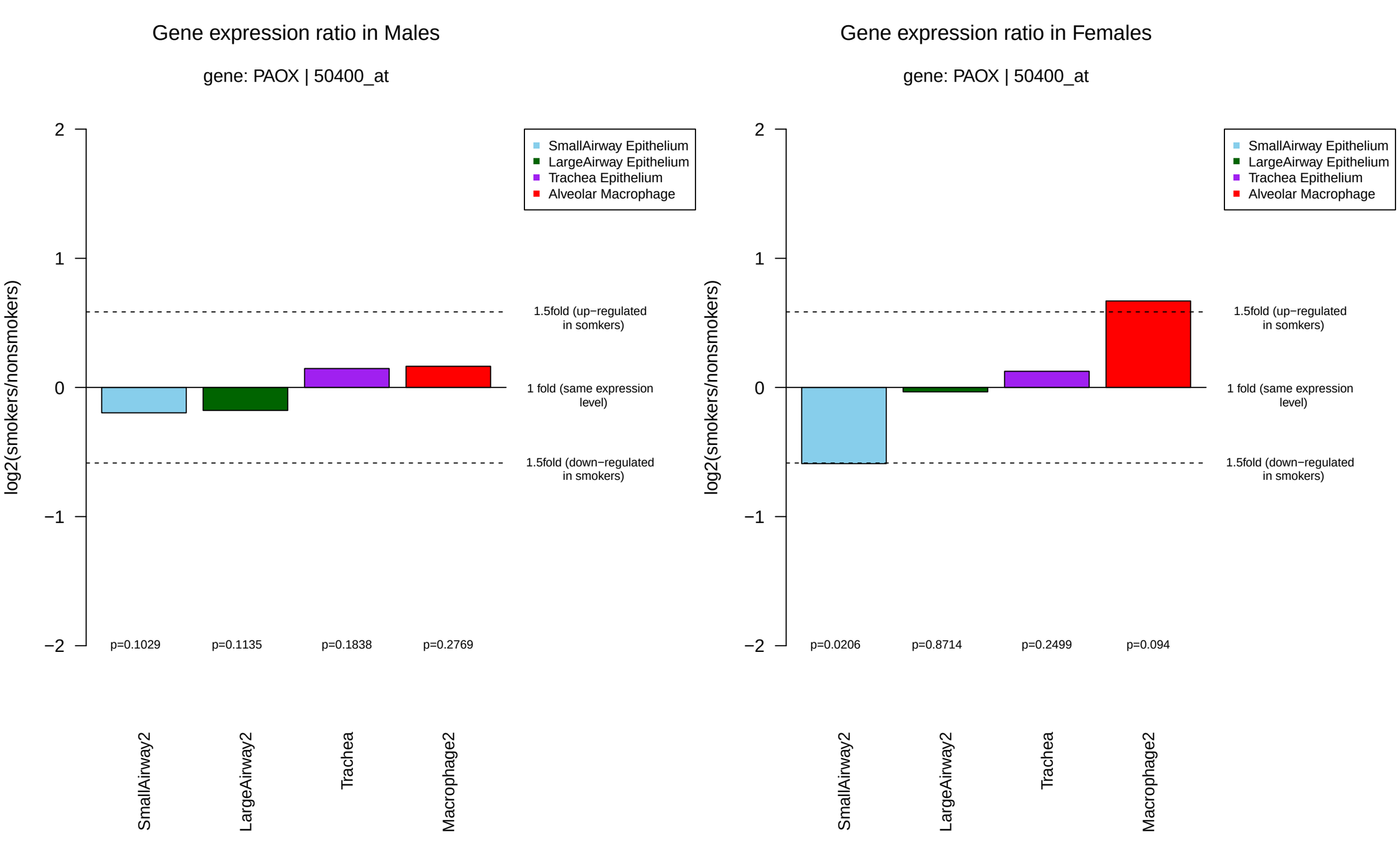
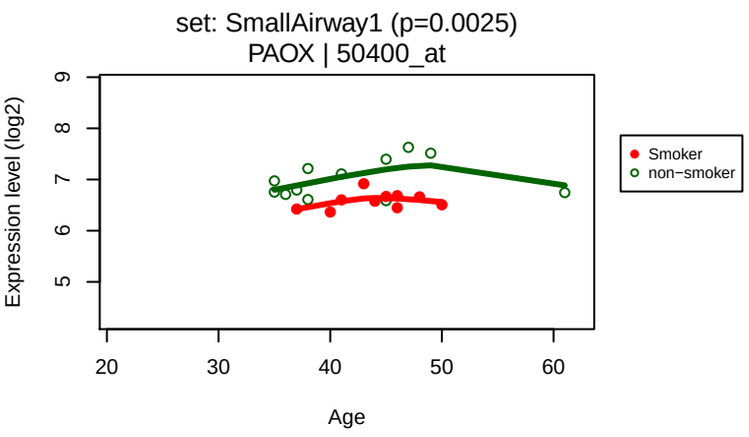
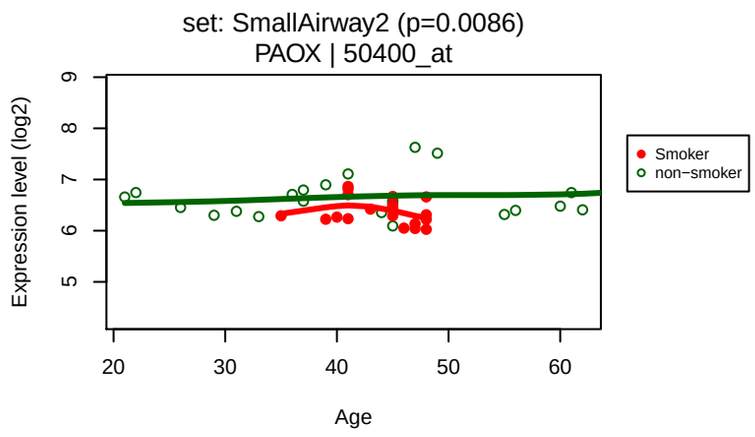
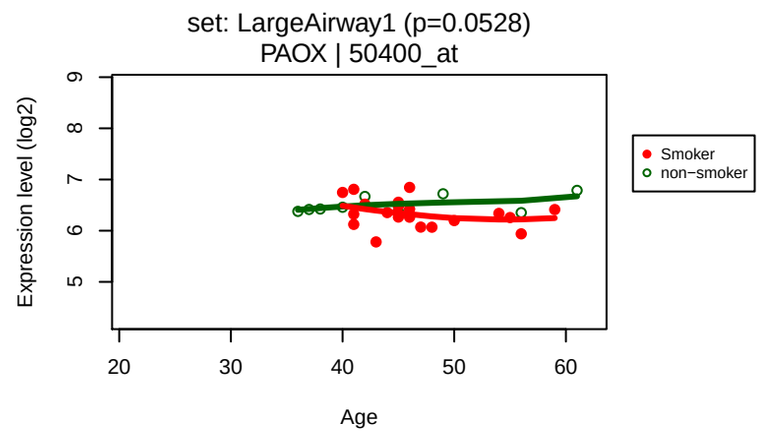

Gene expression of smokers and nonsmokers
gene: PAOX | 50400_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0025
p=0.0086
p=0.0036
p=0.0528
p=0.1888
p=0.0535
p=0.3509
p=0.0234
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

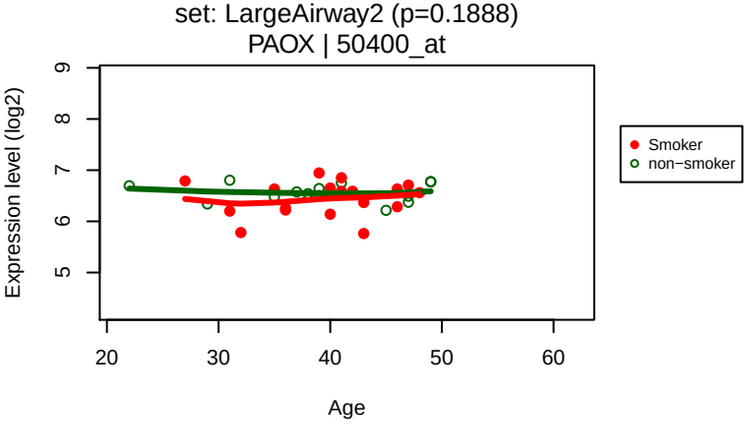

Gene expression ratio (smokers/nonsmokers)
gene: PAOX | 50400_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0025
p=0.0086
p=0.0036
p=0.0528
p=0.1888
p=0.0535
p=0.3509
p=0.0234
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

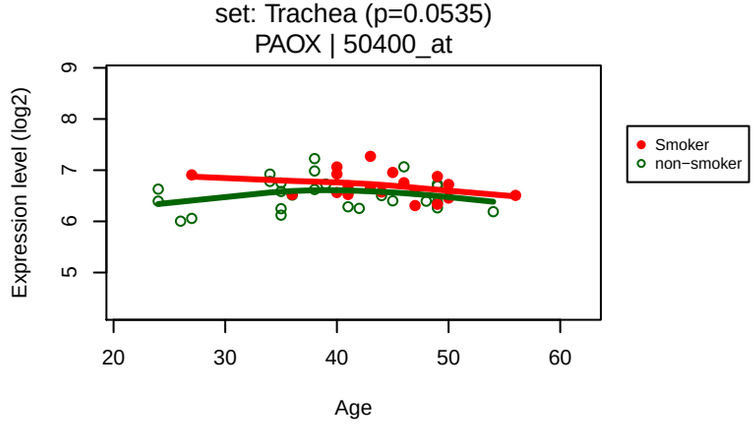

Gene expression ratio in Males
gene: PAOX | 50400_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1029
p=0.1135
p=0.1838
p=0.2769
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAOX | 50400_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0206
p=0.8714
p=0.2499
p=0.094
SmallAirway2
LargeAirway2
Trachea
Macrophage2

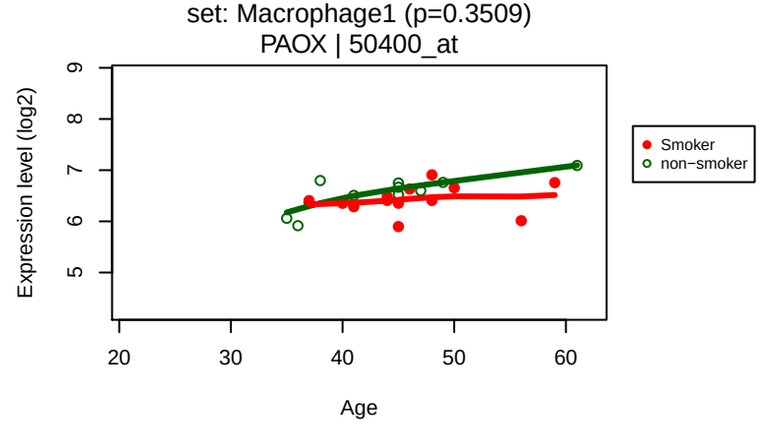

set: SmallAirway1 (p=0.0025)
PAOX | 50400_at
Expression level (log2)
Age
Smoker
non−smoker

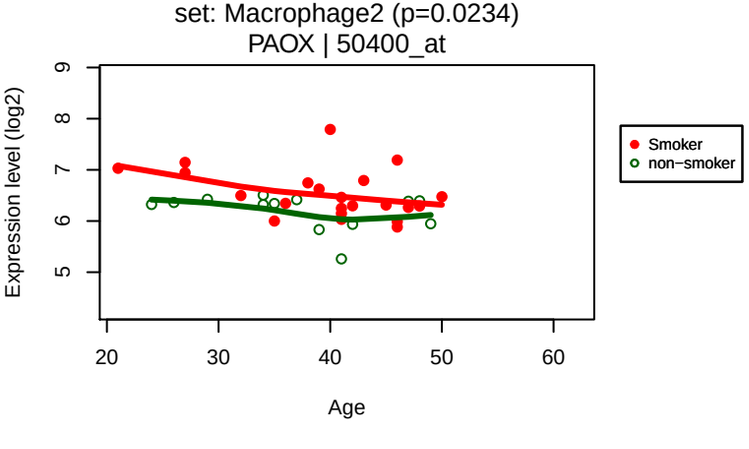

set: SmallAirway2 (p=0.0086)
PAOX | 50400_at
Expression level (log2)
Age
Smoker
non−smoker

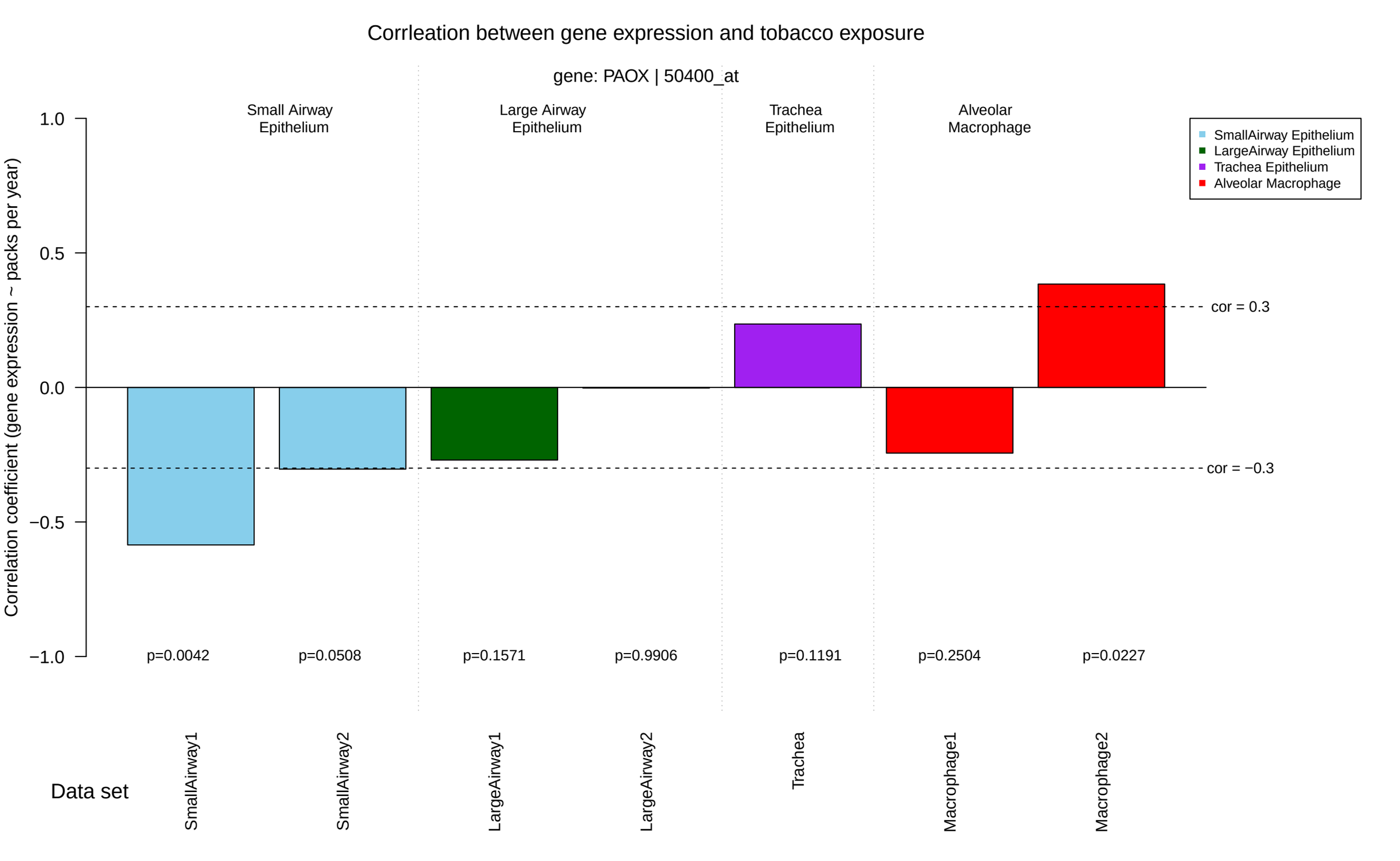

set: LargeAirway1 (p=0.0528)
PAOX | 50400_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1888)
PAOX | 50400_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0535)
PAOX | 50400_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3509)
PAOX | 50400_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0234)
PAOX | 50400_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAOX | 50400_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0042
p=0.0508
p=0.1571
p=0.9906
p=0.1191
p=0.2504
p=0.0227
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2